

The Proton Color Singlet from Symmetric Register Windows: how tetrahedral A_4 symmetry intertwines the color $\mathbb{Z}_3$ at the W_{ABC} junction

Tessera — cobordism subsystem (#398, completing #396)

Abstract

The trivalent junction W_{ABC} (#396) carries the $\mathbb{Z}_3$ shadow of the three-quark color singlet ε_{ijk} that the bipartite merge cannot (#382): three input “windows” A, B, C and one emergent result window R , each a triple of holes, all on one connected 2-frequency geodesic icosahedron (S^2 , 42 vertices, 80 triangles) extruded $\times I$. After #396 the singlet was *reachable* — the input $\rightarrow$ result transport is full-rank with the singlet in its image — but a *natural* quark input reached only ~ 0.74 : the greedily placed windows are geometrically *inequivalent*, so the per-window transport blocks are unrelated matrices. This note records the principled fix and a finding that was not anticipated. We place the four windows as *one orbit of a tetrahedral subgroup* A_4 of the icosahedral rotation group, seated at a system of imprimitivity on the icosahedron’s vertices. We then show that symmetric windows are *necessary but not sufficient*: the singlet emerges only when three conditions hold together — (i) a symmetry-respecting (uniform) metric, (ii) the input in the ω -representation of the window-cycling symmetry g , and (iii) a g -coherent signed read-out. Given these, the transport M *intertwines* the color $\mathbb{Z}_3$, $MP_{\text{in}} = P_{\text{out}}M$, and the ω -representation input transports to the singlet line with overlap 0.999 — the proton, forced by symmetry and never imposed. The residual $\sim 4 \times 10^{-2}$ in the intertwining is a fixed-resolution triangulation artifact (the $\times I$ extrusion is not g -symmetric even though the base surface and metric are); it is metric-independent, and 0.999 rather than an exact 1 is its signature. All numbers are measured through the production C++ `TransportCobordism`.

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1 Where we were

The proton-from-quarks program builds the baryon color singlet by relaxing a Lorentzian Regge cobordism with pinned boundary color states and reading the *emergent* result off the relaxed geometry — nothing is imposed. We write the *color charge* of a window W for the net signed period it carries,

$$\sigma_W = \sum_{\text{holes } h \in W} (\text{signed induced period of } h), \quad (1)$$

the discrete analogue of $\sum_i \psi_i$ on a color triple. Two prior results frame the present one.

The bipartite obstruction (#382). Composing pairwise merges never reaches the singlet: the emergent result stays color-charged ($|\sigma| \approx 0.67$ for the tripartite sequence) at every step. This is representation theory, not a numerical defect: the color singlet ε_{ijk} is the three-index alternating invariant, absent from $\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}$ and first appearing in $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3}$. A baryon from fundamentals is irreducibly an $n = 3$ object.

The tripartite junction (#396). The simplicial resolution is the trivalent junction W_{ABC} : one connected base surface — a 2-frequency geodesic icosahedron (S^2 , 42 vertices, 80 triangles) — minus 12 vertex-disjoint hole triangles grouped into four *windows* of three, A, B, C (the quark inputs) and R (the emergent result), extruded $\times I$ into a 3-complex (**prismCells**). The 12 hole cycles satisfy exactly one relation — the global Stokes constraint below — so $b_1 = 12 - 1 = 11$; the inputs are *independent* cycles (they do not average, escaping the bipartite obstruction), and that single relation is junction charge conservation:

$$\sum_{\text{all holes}} (\text{induced period}) = 0 \quad \implies \quad \sigma_R = -(\sigma_A + \sigma_B + \sigma_C). \quad (2)$$

Confinement is then a property of (2): the four window charges must sum to zero, so an isolated colored input cannot close on the junction; only the color-neutral (singlet) combination does. The bipartite $|\sigma| \approx 0.67$ residual is precisely such a non-confined, charged failure.

The open problem. After #396 the read-out is a linear map $M : \mathbb{C}^9 \rightarrow \mathbb{C}^3$ (nine input color amplitudes, three result amplitudes); it is full-rank ($\text{rank } M = 3$) and the singlet lies in its image, so the singlet is *reachable*. Yet the natural neutral input (three identical color-neutral pairs, read with each hole's own orientation) gives overlap ~ 0.74 . The diagnosis: with the windows placed *greedily* (first twelve vertex-disjoint triangles in face order) the per-window transport blocks M_A, M_B, M_C (the three 3×3 sub-blocks of M) are geometrically inequivalent — $M_A^{-1} M_B$ is a general matrix, not a signed permutation. The input that hits the singlet is therefore geometry-specific and unnatural. The fix is to make the windows *equivalent*, which is the subject of this note.

2 Symmetric placement: the windows as an A_4 orbit

2.1 The symmetry available

The regular icosahedron has rotation group $I \cong A_5$ of order 60; the 2-frequency geodesic subdivision inherits it as a group of simplicial automorphisms permuting the 80 sub-triangles and fixing the metric of the round sphere. A_5 contains tetrahedral subgroups $T \cong A_4$ of order 12. Fix one such A_4 . Because $\gcd(|A_4|, 5) = 1$ and the vertex stabilizer in A_5 is the 5-fold rotation C_5 , A_4 acts *freely and transitively* on the 12 vertices (one orbit of 12). It nevertheless admits a *system of imprimitivity* with four blocks of three,

$$\{2, 8, 10\}, \quad \{1, 4, 7\}, \quad \{0, 6, 9\}, \quad \{3, 5, 11\}, \quad (3)$$

which partition the twelve vertices; A_4 permutes the four blocks by its standard transitive degree-4 action, and each block is fixed setwise by one of A_4 's four C_3 subgroups (its three-fold axis).

2.2 The construction

A *window* is the triple of vertex-disjoint corner sub-triangles seated at one block of (3) and cycled by that block's three-fold axis. Concretely the corner sub-triangle at icosahedron vertex v is $(v, m(v, p), m(v, q))$, where $m(\cdot, \cdot)$ is the edge-midpoint vertex and p, q are two neighbours of v ; the C_3 rotation about the block's axis cycles the three corners, hence the three holes. The four windows W_A, W_B, W_C, W_R are therefore a *single* A_4 orbit (their stabilizer is the block's own C_3 , and $|A_4|/3 = 4$). All 12 holes are vertex-disjoint (the 36 vertices are distinct), and the topology is unchanged: $b_1 = 11$ and the dual complex passes the manifold gate.

Proposition 1 (Symmetric windows). *The four windows in the cobordism's own vertex labels are*

$$\begin{aligned} W_A : & (2, 13, 14) (8, 31, 32) (10, 22, 36) & W_B : & (1, 23, 24) (4, 19, 34) (7, 30, 39) \\ W_C : & (0, 16, 18) (6, 26, 27) (9, 33, 41) & W_R : & (3, 15, 29) (5, 20, 21) (11, 37, 38) \end{aligned}$$

each window a genuine C_3 orbit, all real faces of the base surface, the 36 vertices distinct.

Generated, not hardcoded. The windows are produced *from the symmetry* at build time: four C_3 generators (as 12-vertex permutations), one seed corner per window, and the midpoint lift (a base vertex maps by the permutation; a midpoint $m(p, q)$ maps to $m(p', q')$). This is correct in whatever midpoint numbering the build produces. The point is not pedantic: two independent attempts to *hardcode* the twelve triples both produced tables that were wrong against the actual numbering. Generation is correct-by-construction and is guarded by build-time assertions (each window a C_3 orbit, all holes real faces, 36 distinct vertices). The explicit data are in Appendix A.

3 Necessary but not sufficient: three ingredients

The central empirical finding is that symmetric *windows alone do not produce the singlet*. Three conditions must hold together — a symmetry-respecting metric, the input in the right representation, and a coherent read-out — and all three are phrased in terms of the same window-cycling symmetry, which we name first.

Definition 1 (The window-cycling symmetry g). A_4 acting on the four windows is the alternating group on four points; its three-cycles fix one window and cycle the other three. Let $g \in A_4$ be the element with

$$g : W_R \mapsto W_R, \quad W_A \mapsto W_B \mapsto W_C \mapsto W_A. \quad (4)$$

Such g exists and is unique up to its inverse; on the result window it acts as a C_3 on R 's three holes. *This g is the color $\mathbb{Z}_3$.*

We write P_{in} (resp. P_{out}) for the signed permutation by which g acts on the nine input-hole (resp. three result-hole) period amplitudes; both are made precise in §4.

3.1 (i) A symmetry-respecting metric

The seed metric must be invariant under g . A random “jitter” seed — previously used to lift the degenerate uniform point — breaks the A_4 symmetry: charge conservation (2) degrades from $\sim 10^{-14}$ to ~ 0.07 , and no symmetric input reaches the singlet. The default seed is now the *uniform* metric $l^2 = 1$, which is manifestly g -invariant; on it (2) is exact ($|\sigma_R| \sim 10^{-14}$ for a symmetric neutral input). The mutual-information (van Raamsdonk) seed, `setEntangledMetric` — entanglement-derived edge lengths that are constant on g -orbits, hence g -invariant — works equally well.

Crucially, the singlet is read from the *transport* at the symmetric seed, not from a relaxation: the intertwining of §4 holds already at iteration zero. Relaxation still runs and is well-behaved (60 iterations reduce the stationarity residual from 204 to 8.4), but it is not what produces the singlet.

3.2 (ii) The ω -representation input

The “natural symmetric quark input” is *not* $[1, \omega, \omega^2]$ placed identically on A, B, C (that gives only ~ 0.48). It is the input that transforms in the ω -representation of g :

$$P_{\text{in}} \phi = \omega \phi, \quad \omega = e^{2\pi i/3}. \quad (5)$$

Physically: the three quark colors carry the correct relative $\mathbb{Z}_3$ phase under the geometric cycling, not the same vector repeated. The input representation is fixed by g — a property of the construction — *before* any output is computed; the theorem’s content (§4) is that the transport carries it to the singlet line rather than scrambling it, as the failed rows of §5 show it does for non- ω -rep inputs.

3.3 (iii) A g -coherent signed read-out

The period of a 1-cochain around a hole triangle is orientation-dependent. The read-out must use orientations coherent under g : the period of each hole is taken in the orientation propagated from a chosen base hole by g , i.e. multiplied by the orientation parity ε of g on it (the corresponding entry of P). Equivalently, on the three holes of R the read-out signs are the entries of P_{out} . The default per-window covector is *not* g -coherent — its signs are inconsistent across windows ($[-1, -1, 1], [1, -1, -1], [1, -1, 1]$) — and with raw sorted periods the symmetry is broken in the read-out and the singlet is missed.

4 The intertwining theorem

Let $M : \mathbb{C}^9 \rightarrow \mathbb{C}^3$ be the input→result transport: M sends the nine pinned input-hole periods to the three result-hole periods of the carried harmonic. M is linear because the carried-representative map is linear.

Lemma 1 (Equivariance on the base surface). *On the base surface (the holed geodesic sphere), g is a metric-preserving simplicial automorphism. It acts on hole-period vectors by a signed permutation P , the sign on a hole being the orientation parity of g on it; $P = P_{\text{in}} \oplus P_{\text{out}}$ because g maps input holes to input holes and result holes to result holes. The base-surface carried map commutes with g :*

$$M P_{\text{in}} = P_{\text{out}} M. \quad (6)$$

Proof. g preserves edge lengths, so it preserves the discrete Hodge inner products and commutes with the coboundary and Hodge Laplacian; the least-norm harmonic realizing a set of pinned periods is carried to the harmonic realizing the g -transported periods. A hole (a, b, c) (oriented, sorted) has period $\psi_{ab} + \psi_{bc} - \psi_{ac}$, invariant under cyclic relabelling and negated under a transposition; g sends $(a, b, c) \mapsto (ga, gb, gc)$, whose sorted form differs by a permutation of parity $\varepsilon \in \{\pm 1\}$, so the period maps with sign ε . Collecting the signs gives $P = P_{\text{in}} \oplus P_{\text{out}}$ and (6). $\square$

Theorem 1 (The ω -input transports to the singlet). *Assume the exact equivariance (6). Then P_{out} is a signed 3-cycle with eigenvalues $\{1, \omega, \omega^2\}$, and its ω -eigenspace is the one-dimensional color singlet line $\mathbb{C}[1, \omega, \omega^2]$ in R 's hole basis. For any input ϕ with $P_{\text{in}}\phi = \omega\phi$, the result $M\phi$ lies in that singlet line. Moreover M is rank 3 with $\phi \notin \ker M$ (so $M\phi \neq 0$), and the ω -eigenspace of P_{in} is three-dimensional, so a three-parameter family of symmetric inputs all transport to the singlet.*

Proof. g cycles R 's three holes, so P_{out} is a signed 3-cycle; the product of its three signs is $+1$ (verified: the eigenvalue angles are $\{0, \pm 2\pi/3\}$), hence its eigenvalues are the cube roots of unity, each eigenspace one-dimensional. The ω -eigenvector is the discrete Fourier mode $[1, \omega, \omega^2]$ — the color singlet. For $P_{\text{in}}\phi = \omega\phi$, equation (6) gives $P_{\text{out}}(M\phi) = M(P_{\text{in}}\phi) = \omega(M\phi)$, so $M\phi$ is an ω -eigenvector of P_{out} , i.e. $M\phi \in \mathbb{C}[1, \omega, \omega^2]$; and $M\phi \neq 0$ since M is rank 3 and ϕ is not in its kernel. Finally P_{in} is three disjoint signed 3-cycles (each an A -hole, its B -image, its C -image), each with sign-product $+1$, so ω occurs with multiplicity three. $\square$

Remark 1 (The measured object is equivariant only up to a triangulation residual). g is a metric-preserving automorphism of the base surface, but the cobordism is the base $\times I$, and the prism tetrahedralization's diagonal choices are not g -invariant — g permutes vertices but not simplices. Hence the carried map *through* the 3-complex is only approximately equivariant, $\|MP_{\text{in}} - P_{\text{out}}M\| / \|M\| \approx 4.3 \times 10^{-2}$ (§5), and the ω -input transports to the singlet line up to that residual — overlap 0.999 rather than exactly 1. The residual is metric-independent (§5); a g -symmetric subdivision of the extrusion would remove it.

Remark 2 (Manifest $\mathbb{Z}_3$, and the relation to S_3). What is geometrically realized is the color $\mathbb{Z}_3$: g is its generator, the transport intertwines it, and the singlet is its ω -Fourier mode — the $\mathbb{Z}_3$ mode that survives restricting the totally antisymmetric ε_{ijk} to the cyclic subgroup. The full color Weyl group is S_3 ; its $\mathbb{Z}_3$ cyclic part is the one made manifest by the tetrahedral geometry, and the conjugate symmetric input ($P_{\text{in}}\phi = \omega^2\phi$) lands on the conjugate singlet $[1, \omega^2, \omega]$. The singlet is thus not a tuned target but the forced image of an entire eigenspace — the sense in which it is emergent rather than imposed.

5 Numerical results

All figures are measured through the production C++ `TransportCobordism` with the symmetric `TripartiteRegisterTopology`; the symmetry g and the signed reps $P_{\text{in}}, P_{\text{out}}$ are reconstructed from the icosahedral generators applied to the cobordism’s *own* window labels (never Python-reconstructed holes). The figure of merit is the normalized overlap

$$\text{overlap} = \frac{|\langle v_R, s \rangle|}{\|v_R\| \|s\|}, \quad s = [1, \omega, \omega^2], \quad (7)$$

where v_R is the carried result-hole period vector (read from R after the carry, never pinned).

Placement / metric / input	overlap with s	note
greedy windows, jitter, natural neutral input	~ 0.74	windows inequivalent
symmetric windows, uniform, $[1, \omega, \omega^2]$ identical on A, B, C	0.48	wrong input (not ω -rep)
symmetric windows, uniform, naive neutral, raw read	0.65	wrong (non-coherent) read-out
symmetric windows, uniform, ω-rep input	0.999	the proton

The structural and symmetry measurements:

- topology: $b_1 = 11$, dual complex valid (a manifold, no weld); metric uniform.
- P_{out} eigenvalues $\{1, \omega, \omega^2\}$ (angles $\{0, \pm 2\pi/3\}$): the color $\mathbb{Z}_3$.
- intertwining $\|MP_{\text{in}} - P_{\text{out}}M\| / \|M\| = 4.26 \times 10^{-2}$.
- the three ω -eigenvectors of P_{in} map to the singlet with overlaps 0.9991, 0.9998, 0.9991; the small spread is the fixed-resolution triangulation artifact (basis-dependent), with 0.999 the reported floor.
- charge conservation (2) exact (10^{-14}) for a symmetric neutral input on the uniform metric; the singlet is traceless, $1 + \omega + \omega^2 = 0$.

The residual is a triangulation artifact, not symmetry breaking. The intertwining is not exactly zero, and the cause is *not* the metric: replacing the uniform seed by a g -invariant *non-degenerate* metric (edge lengths constant on g -orbits) gives the same residual, 4.19×10^{-2} versus 4.26×10^{-2} . The base surface, its holes, and the metric are all g -symmetric; what is not is the prism tetrahedralization (the $\times I$ extrusion’s diagonal choices), so the discrete carried map is only approximately equivariant at this fixed resolution (Remark 1). The 0.999 overlap, not the exact 1, is the signature of that artifact; a g -symmetric subdivision would remove it. This is the conclusive figure that the symmetry is respected. *That subdivision has since been built* (the symmetric apex interior, #413): replacing the prism extrusion with a label-independent stacking — each base triangle coning to a face-apex, the gap octahedra split along the canonical dual edge $f_1 f_2$, so no vertex-label sort chooses a diagonal — drives the residual from 4.26×10^{-2} to 4.5×10^{-14} (machine zero) and the singlet overlap to 1.000000, confirming the artifact diagnosis exactly.

6 Implementation

The change is confined to `TripartiteRegisterTopology`.

- `geodesicTwoSphere` now returns its edge-midpoint map alongside the faces, so the window generator can lift the C_3 permutations.
- `symmetricWindows` generates the four windows from the four C_3 generators, a seed corner per window, and the midpoint lift, with the build-time assertions of Proposition 1; it replaces the greedy twelve-hole pick.
- the default metric seed is uniform $l^2 = 1$ (the jitter seed is removed); the van Raamsdonk seed (`setEntangledMetric`) remains as a g -invariant alternative.

The result block R still *emerges* — it is never pinned or hand-placed: the read-out pins the three inputs on A, B, C and returns R to be read off the transport after the carry. Validation reconstructs g and $P_{\text{in}}, P_{\text{out}}$ from the cobordism’s own labels and checks Theorem 1 end-to-end. A worked demonstration is `examples/cobordism/proton_tripartite_junction.py`.

7 Discussion

The proton singlet is now *forced by symmetry*. With symmetric windows and a symmetric metric the transport intertwines the color $\mathbb{Z}_3$; the color-symmetric quark input is, by definition, the ω -representation; and any such input lands on the unique ω -eigenvector of the result, the singlet $[1, \omega, \omega^2]$. Nothing about the singlet is inserted: it is the image of an entire eigenspace, and a three-parameter family of symmetric inputs all reach it. The earlier 0.74 ceiling was not a limitation of the geometry but of *inequivalent* windows; the earlier 0.48 and 0.65 “failures” were the wrong input and the wrong read-out, not the wrong construction.

Two honest caveats. First, the geometrically realized symmetry is $\mathbb{Z}_3$, the cyclic subgroup of the color Weyl group S_3 ; the cyclic singlet is what is forced. Second, the intertwining is exact only up to a fixed-resolution triangulation artifact ($\sim 4 \times 10^{-2}$), so the measured overlap is 0.999 rather than 1 (Remark 1).

The path forward (#380) reads emergent observables off the relaxed singlet geometry — net charge, a radius, the mass ratio — and follows the photon (a worldline whose l^2 relaxes through null) into re-absorption.

8 Conclusion

The trivalent junction with *symmetric* register windows resolves the last gap in building the proton from quarks. Placing the four windows as one tetrahedral A_4 orbit makes them equivalent; on a symmetry-respecting metric the input→result transport intertwines the color $\mathbb{Z}_3$; and the natural color-symmetric input then transports to the color singlet with overlap 0.999. Symmetric windows are necessary but not sufficient — the metric, the input representation, and the read-out must all respect the same symmetry — and when they do, the proton’s color singlet is not assembled but *compelled*.

A Explicit data

The four C_3 generators as 12-vertex permutations (image of vertex i at position i), each cycling its window’s vertex block:

$$\begin{aligned}
W_A (2 \rightarrow 8 \rightarrow 10) : & \quad [4, 3, 8, 9, 5, 0, 7, 11, 10, 1, 2, 6] \\
W_B (1 \rightarrow 4 \rightarrow 7) : & \quad [3, 4, 0, 2, 7, 8, 5, 1, 6, 11, 9, 10] \\
W_C (0 \rightarrow 6 \rightarrow 9) : & \quad [6, 10, 11, 7, 2, 1, 9, 8, 3, 0, 5, 4] \\
W_R (3 \rightarrow 5 \rightarrow 11) : & \quad [10, 6, 1, 5, 9, 11, 2, 0, 4, 8, 7, 3]
\end{aligned}$$

The seed corner of window w is $(v, m(v, n_1), m(v, n_2))$ with $(v; n_1, n_2)$ equal to

$$(2; 0, 1), \quad (1; 6, 10), \quad (0; 3, 4), \quad (3; 2, 7)$$

respectively; the window is the C_3 orbit of its seed. The window-cycling element g of Definition 1 is the A_4 rotation whose induced window permutation is $W_A \rightarrow W_B \rightarrow W_C \rightarrow W_A$ with W_R fixed; it acts on R 's three holes as a 3-cycle, giving P_{out} its eigenvalues $\{1, \omega, \omega^2\}$.